

Project Initialization and Planning Phase

Date	8 July 2026
Team ID	
Project Title	Credit Card Approval Prediction System
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) Report:

The proposed solution aims to automate the credit card approval process using machine learning techniques, improving both efficiency and prediction accuracy. The system analyzes applicants' financial and demographic information to determine whether a credit card application is likely to be approved or rejected. By reducing manual verification, the project helps banks make faster, more consistent, and data-driven decisions through a user-friendly web application.

Project Overview	
Objective	To develop a machine learning-based credit card approval prediction system that provides fast, accurate, and reliable approval or rejection decisions using applicant data.
Scope	The project covers data preprocessing, feature engineering, machine learning model training, model evaluation, and deployment of a Flask web application for real-time credit card approval prediction.
Problem Statement	
Description	Manual credit card application verification is time-consuming, prone to human error, and inefficient when processing a large number of applications.
Impact	Automating the approval process improves operational efficiency, reduces processing time, minimizes human errors, and provides consistent prediction results for banks and customers.

Proposed Solution

Section	Details
Approach	Use machine learning algorithms to analyze applicant information such as income, employment duration, education level, and credit history to predict credit card approval or rejection through a Flask web application.
Key Features	• Machine learning-based credit card approval prediction. • Data preprocessing and feature engineering. • Comparison of Logistic Regression, Decision Tree, Random Forest, and XGBoost models. • Automatic selection of the best-performing model. • Real-time prediction using a Flask web application. • Cloud deployment using IBM Watson Machine Learning.

Resource Requirements

Hardware

Resource Type	Description	Specification / Allocation
Computing Resources	CPU for model training and prediction	Intel Core i5/i7 or equivalent
Memory	RAM	8 GB or higher
Storage	Dataset, trained models, and project files	10 GB or more SSD

Software

Resource Type	Description	Specification / Allocation
Framework	Web Application Framework	Flask
Libraries	Machine Learning & Data Processing	Scikit-learn, XGBoost, Pandas, NumPy, Matplotlib, Seaborn, Joblib
Development Environment	IDE	Jupyter Notebook, VS Code, PyCharm

Data

Resource Type	Description	Specification / Allocation
Data	Source, Format	Kaggle Credit Card Approval Dataset (application_record.csv and credit_record.csv) in CSV format